

3.1: Bisection Method

5) Determine the formula

$$n \geq \frac{\log(b_0 - a_0) - \log(\epsilon) - \log(a_0)}{\log 2} - 1$$

involving $b_0 - a_0$ and ϵ for the number of steps that must be taken in the bisection method to guarantee that $|r - c_n| \leq \epsilon$.

Solution:

From the Theorem on Bisection Method, let $[a_i, b_i]$ denote the intervals where $0 \leq n$ where the limits $\lim_{n \rightarrow \infty} a_n$ and $\lim_{n \rightarrow \infty} b_n$ exists, are equal, and represent a zero of F . With $r = \lim_{n \rightarrow \infty} c_n$ and $c_n = \frac{1}{2}(a_n + b_n)$, then

$$|r - c_n| \leq 2^{-(n+1)}(b_0 - a_0)$$

Furthermore, we can express the relative precision as

$$\frac{|r - c_n|}{|r|} \leq \epsilon.$$

Assume $a_0 > 0$, then with $0 < a_0 \leq r$ we can algebraically determine the formula as follows:

$$\begin{aligned} \epsilon &\geq \frac{|r - c_n|}{a_0} \\ &\geq \frac{2^{-(n+1)}(b_0 - a_0)}{a_0} \\ a_0 \epsilon &\geq 2^{-(n+1)}(b_0 - a_0) \\ \frac{a_0 \epsilon}{(b_0 - a_0)} &\geq 2^{-(n+1)} \\ \log\left(\frac{a_0 \epsilon}{(b_0 - a_0)}\right) &\geq \log\left(2^{-(n+1)}\right) \\ \frac{\log\left(\frac{a_0 \epsilon}{(b_0 - a_0)}\right)}{\log(2)} &\geq -(n+1) \\ n &\geq \frac{\log\left(\frac{a_0 \epsilon}{(b_0 - a_0)}\right)}{\log(2)} - 1 \\ &= \frac{\log(b_0 - a_0) - \log(\epsilon) - \log(a_0)}{\log(2)} - 1 \end{aligned}$$

□

3.2: Newton's Method

- 3) Prove that if Newton's method is used on a function f for which f'' is continuous and $f(r) = 0 \neq f'(r)$, then $\lim_{n \rightarrow \infty} e_{n+1}e_n^{-2}$ exists and equals $\frac{f''(r)}{2f'(r)}$. How can this fact be used in a program to test whether convergence is quadratic?

Solution:

Recall from the text the following formulas will be reference for this proof:

$$\begin{aligned} e_n &= x_n - r \\ e_{n+1} &= x_{n+1} - r \\ &= x_n - \frac{f(x_n)}{f'(x_n)} - r = e_n - \frac{f(x_n)}{f'(x_n)} \\ &= \frac{e_n f'(x_n) - f(x_n)}{f'(x_n)} \\ e_{n+1} &\approx \frac{f''(r)}{2f'(r)} e_n^2 = C e_n^2 \end{aligned}$$

Let f be a function for which f'' is continuous and $f(r) = 0 \neq f'(r)$, then using the formula (3) (the last line above) as follows:

$$\begin{aligned} e_{n+1} &= e_n^2 \cdot \left(\frac{f''(\xi_n)}{2f'(x_n)} \right) \\ \implies e_{n+1}e_n^{-2} &= \frac{f''(\xi_n)}{2f'(x_n)} \end{aligned}$$

where ξ_n is between x_n and r . Suppose we are working in the region of convergence, then the following holds:

$$\begin{aligned} \lim_{n \rightarrow \infty} x_n &= r \\ \lim_{n \rightarrow \infty} e_n &= 0 \\ \implies \lim_{n \rightarrow \infty} f'(\xi_n) &= f'(r) \\ \implies \lim_{n \rightarrow \infty} f''(\xi_n) &= f''(r) \end{aligned}$$

Next, taking the limit of the manipulated formula (3) from above to yield the following:

$$\begin{aligned} \lim_{n \rightarrow \infty} e_{n+1}e_n^{-2} &= \lim_{n \rightarrow \infty} \frac{f''(\xi_n)}{2f'(x_n)} \\ &= \frac{f''(r)}{2f'(r)} \end{aligned}$$

as desired. After Newton's Method has been implemented to yield root $= r$, then each $x_1, x_2, \dots, x_n$ where $x_n = r$ to machine precision would also be derived. Hence the sequence $e_1, e_2, e_3, \dots, e_n$ can be computed from the definition

$$e_k = x_k - r.$$

This further allows the calculation and convergence of:

$$\frac{e_{k+1}}{e_k^2} \implies \frac{f''(r)}{2f'(r)}$$

□

4) (**Steffensen's Method**) Consider the iteration formula

$$x_{n+1} = x_n - \frac{f(x_n)}{g(x_n)}$$

where

$$g(x) = \frac{f(x + f(x)) - f(x)}{f(x)}$$

Show that this is quadratically convergent, under suitable hypotheses.

Solution:

Given the iteration formula, consider the Taylor series expansion on the numerator component of $g(x)$:

$$\begin{aligned} f(x+h) &= f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \cdots \\ \implies f(x+f(x)) &= f(x) + f(x)f'(x) + \frac{f(x)^2}{2}f''(x) + \cdots \\ g(x) &= \frac{f(x+f(x)) - f(x)}{f(x)} = f'(x) + \frac{f(x)f''(x)}{2} + \cdots \end{aligned}$$

Hence, if f is small, then

$$\frac{f(x+f(x)) - f(x)}{f(x)} \approx f'(x),$$

this consequently leads to

$$g(x) \approx f'(x).$$

Next, to show that $g(x)$ is quadratically convergent, consider the formulas from #3 above (and from the text) and the given iteration formula:

$$\begin{aligned} e_{n+1} &= x_{n+1} - r \\ &= x_n - \frac{f(x_n)}{g(x_n)} \\ &= e_n - \frac{f}{g} \\ &= \frac{e_n g - f}{g} \implies f - e_n g = -g e_{n+1} \\ 0 &= f(r) \\ &= f(x_n - e_n) \quad (\text{since } e_n = x_n - r) \\ &= f(x_n) - e_n f'(x_n) + \frac{e_n^2 f''(\xi_n)}{2} \quad (\text{by the Taylor Expansion above}) \\ f - e_n g &= -\frac{1}{2} e_n^2 f'' \implies e_{n+1} = \frac{e_n^2 f''}{2g} \end{aligned}$$

Therefore,

$$e_{n+1} = \frac{e_n^2}{2} \left(\frac{f''}{g} \right)$$

and the given iteration formula is quadratically convergent. $\square$

- 7) Define $x_0 = 0$ and $x_{n+1} = x_n - \frac{\tan(x_n) - 1}{\sec^2(x_n)}$. What is $\lim_{n \rightarrow \infty} x_n$ in this example? Relate this to Newton's Method.

Solution:

Let $f(x) = \tan(x) - 1$ then taking the derivative to yield

$$f'(x) = \sec^2 x.$$

Then the given sequence is an application of Newton's Method to determine a zero of $f(x)$. The root can be found analytically as $x = \tan^{-1}(1) = 0.785398$. Thus,

$$\lim_{n \rightarrow \infty} x_n = 0.785398$$

□

3.4: Fixed Points and Functional Iteration

- 1) If F is contractive from $[a, b]$ to $[a, b]$ and $x_{n+1} = F(x_n)$ with $x_0 \in [a, b]$, then $|x_n - s| \leq C\lambda^n$ for an appropriate C . Prove this and give an upper bound for C . Here s is the fixed point of F .

Solution:

Suppose F is contractive from $[a, b]$ to $[a, b]$. Then by definition

$$|F(x) - F(y)| \leq \lambda|x - y|$$

for $\lambda < 1$. In the (text) proof of the Contraction Mapping Theorem, it was shown that

$$|x_n - x_m| \leq \lambda^n |x_1 - x_0| (1 - \lambda)^{-1}.$$

This holds for $n \geq m \geq N$. Thus, for $n \geq n$

$$\begin{aligned} |x_n - x_N| &\leq \lambda^N |x_1 - x_0| (1 - \lambda)^{-1} \\ \implies \lim_{n \rightarrow \infty} x_n &= \xi \\ \implies |\xi - x_N| &\leq \lambda^N |x_1 - x_0| (1 - \lambda)^{-1} \\ &\leq \lambda^N |b - a| (1 - \lambda)^{-1} \\ &= \frac{|b - a|}{(1 - \lambda)^{-1}} \end{aligned}$$

Thus, let $C = \frac{|b-a|}{1-\lambda}$ then

$$|x_n - \xi| \leq C\lambda^n$$

for any $n \geq N$. □

- 2) Prove that if $F : [a, b] \rightarrow \mathbb{R}$, if F' is continuous, and if $|F'(x)| < 1$ on $[a, b]$, then F is a contraction. Does F necessarily have a fixed point?

Solution:

To prove F is a contraction, then by definition (as in #1)

$$|F(x) - F(y)| \leq \lambda|x - y|$$

for $\lambda < 1$. Suppose $F : [a, b] \rightarrow \mathbb{R}$ F' is continuous, and $|F'(x)| < 1$ on $[a, b]$ and let x_n be a sequence, then by (the limit) definition of derivatives

$$\begin{aligned} F'(x_n) &= \frac{F(x_n) - F(x_{n-1})}{x_n - x_{n-1}} \\ |F'(x_n)| < 1 &\implies \frac{F(x_n) - F(x_{n-1})}{x_n - x_{n-1}} < 1 \\ |F(x_n) - F(x_{n-1})| &< (1) |x_n - x_{n-1}| \end{aligned}$$

Thus, by definition, $F(x)$ is a contraction. Also, by the **Contractive Mapping Theorem**, since F is contractive, then F has a fixed point. □

- 4) Show that these functions are contractive on the indicated intervals. Determine the best values of λ in the following equation:

$$|F(x) - F(y)| \leq \lambda |x - y|$$

Solution:

- (a) $(1 + x^2)^{-1}$ on an arbitrary interval.

Let $F(x) = (1 + x^2)^{-1} = \frac{1}{(1+x^2)}$, then differentiating to yield the following:

$$\begin{aligned} F'(x) &= -(1 + x^2)^{-2}(2x) \\ &= -2x(1 + x^2)^{-2} \\ &= \frac{-2x}{(1 + x^2)^2} \end{aligned}$$

On any arbitrary interval (and graphing using Desmos.com), the maximum value of $|F'(x)|$ occurs at $x = \frac{1}{\sqrt{3}}$, so

$$F\left(\frac{1}{\sqrt{3}}\right) = \left(1 + \left(\frac{1}{\sqrt{3}}\right)^2\right)^{-1} = 0.6495.$$

Thus,

$$\begin{aligned} |F(x) - F(y)| \leq \lambda |x - y| &\implies \frac{|F'(x) - F(y)|}{|x - y|} \leq \lambda \\ |F'(x)| &\leq 0.6495 \\ \implies \lambda &= 0.6495 \end{aligned}$$

□

- 19) If the method of functional iteration is applied to the function $F(x) = 2 + (x - 2)^4$, starting at $x = 2.5$, what order of convergence results? Find the range of starting values for which this functional iteration converges. Note that 2 is a fixed point.

Solution:

Let $F(x) = 2 + (x - 2)^4$ starting with $x = 2.5$, then using the method of functional iteration to yield the following derivatives:

$$\begin{aligned} F'(x) &= 4(x - 2)^3 \implies F'(x) = 0 \\ F''(x) &= 12(x - 2)^2 \implies F''(2) = 0 \\ F'''(x) &= 24(x - 2)^1 \implies F'''(2) = 0 \\ F^{(4)}(x) &= 24 \implies F^{(4)}(2) = 24 \neq 0 \end{aligned}$$

Thus, the order of convergence is 4. Further, to determine the range of starting values for which this function iteration converges:

$$\begin{aligned} |F'(x)| < 1 &\implies |4(x - 2)^3| < 1 \\ 4|x - 2|^3 < 1 &\implies |x - 2|^3 < \frac{1}{4} = 0.25 \\ \implies |x - 2| &< \frac{1}{\sqrt[3]{4}} = 0.63 \end{aligned}$$

Thus, the range is $\left(2 - \frac{1}{\sqrt[3]{4}}, 2 + \frac{1}{\sqrt[3]{4}}\right) \approx (1.37, 2.63)$

□